Q. No	Sub Q.N.	Answer	Marking Scheme
1.		**Attempt any FIVE of the following:**	**10**
	a)	**Differentiate between Multi programmed and Multitasking operating system (Any two points)**	**2M**
	Ans.	(see table below)	***Any two relevant points, 1M each***

Features	**Multiprogramming**	**Multitasking**
Basic	It allows multiple programs to utilize the CPU simultaneously.	A supplementary of the multiprogramming system also allows for user interaction.
Mechanism	Based on the context switching mechanism.	Based on the time-sharing mechanism.
Objective	It is useful for reducing/decreasing	It is useful for running multiple processes at the

	CPU idle time and increasing throughput as much as possible.	same time, effectively increasing CPU and system throughput.
Execution	When one job or process completes its execution or switches to an I/O task in a multi-programmed system, the system momentarily suspends that process. It selects another process from the process scheduling pool (waiting queue) to run.	In a multiprocessing system, multiple processes can operate simultaneously by allocating the CPU for a fixed amount of time.
CPU Switching	In a multiuser environment, the CPU switches between programs/processes quickly.	In a single-user environment, the CPU switches between the processes of various programs.
Timing	It takes maximum time to execute the process.	It takes minimum time to execute the process.

with I/O resources.

Differentiate between paging and segmentation. **2M**

Any two relevant differences – 1M each

Parameters	Paging	Segmentation
Individual Memory	In Paging, we break a process address space into blocks known as pages.	In the case of Segmentation, we break a process address space into blocks known as sections/segments.
Memory Size	The pages are blocks of fixed size.	The sections/segments are blocks of varying sizes.
Accountability	The OS divides the available memory into individual pages.	The compiler mainly calculates the size of individual segments, their actual address as well as virtual address.
Speed	This technique is comparatively much faster in accessing memory.	This technique is comparatively much slower in accessing memory than Paging.
Size	The available memory determines the individual page sizes.	The user determines the individual segment sizes.
Fragmentation	The Paging technique may underutilize some of the pages- thus leading to internal fragmentation.	The Segmentation technique may not use some of the memory blocks at all. Thus, it may lead to external fragmentation.
Logical Address	A logical address divides into page offset and page number in the case of Paging.	A logical address divides into section offset and section number in the case of Segmentation.
Data Storage	In the case of Paging, the page table leads to the storage of the page data.	In the case of Segmentation, the segmentation table leads to the storage of the segmentation data.

Ans.

Parameter	Command Line Interface(CLI)	Graphic User Interface(GUI)
Definition	Interaction is by typing commands	Interaction with devices is by graphics and visual components and icons
Understanding	Commands need to be memorized	Visual indicators and icons are easy to understand
Memory	Less memory is required for storage	More memory is required as visual components are involved.
Working Speed	Use of keyboard for commands makes CLI quicker.	Use of mouse for interaction makes it slow
Resources used	Only keyboard	Mouse and keyboard both can be used
Accuracy	High	Comparatively low
Flexibility	Command line interface does not change, remains same over time	Structure and design can change with updates

Any four points 1M each

b)	**Write any four systems call related to file management**	**4M**

c)	Compare between Long term and short term scheduler. (Any four points)	4M
Ans.	(see table below)	*Any four points 1M each*

Sr. No	Long Term Scheduler	Short Term Scheduler
1	It is job scheduler	It is CPU scheduler
2	It selects processes from job pool and loads them into memory for execution	It selects processes from ready queue which are ready to execute and allocates CPU to one of them
3	Access job pool and ready queue	Access ready queue and CPU
4	It executes much less frequently. It executes when memory has space to accommodate new process.	It executes frequently. It executes when CPU is available for allocation
5	Speed is less than short term scheduler	Speed is fast
6	It controls the degree of multiprogramming	It provides lesser control over degree of multiprogramming
7	It chooses a good process that is a mix-up of input/output bound and CPU bound.	It chooses a new process for a processor quite frequently.

4.		Attempt any THREE of the following:	12
	a)	**Compare between Time sharing operating system and multiprogramming operative system.**	4M

Ans.

Time sharing operating system	Multiprogramming operating system
In this time sharing Operating system many users/processes are allocated with computer resources in respective time slots.	Multiprogramming operating system allows to execute multiple processes by monitoring their process states and switching in between processes.
Processors time is shared with multiple users that's why it is called as time sharing operating system.	Processor and memory underutilization problem is resolved and multiple programs runs on CPU that's why it is called multiprogramming.
In this process, two or more users can use a processor in their terminal.	In this, the process can be executed by a single processor.
Time sharing OS has fixed time slice.	Multi-programming OS has no fixed time slice.
Time sharing system minimizes response time.	Multiprogramming system maximizes processor use.
Example: Windows NT.	Example: Mac OS.

Any 4 correct points 1M each

	d)	Solve given problem by using SJF and FCFS scheduling algorithm using Gantt chart. Calculate the average waiting time for each algorithm	4M

Process	Burst time (in ms)
P1	9
P2	7
P3	3
P4	7

Ans.

Gantt Chart SJF

P3	P2	P4	P1
0 – 3	3 – 10	10 – 17	17 – 26

Waiting Time
P1=17
P2=3
P3=0
P4=10

Average waiting time=Waiting time of all processes / Number of processes

=(17+3+0+10) /4
=30/4
=7.5 milliseconds (ms)

Gantt Chart FCFS

P1	P2	P3	P4
0 – 9	9 – 16	16 – 19	19 – 26

Waiting Time
P1=0
P2=9
P3=16
P4=19

Average waiting time=Waiting time of all processes / Number of processes

=(0+9+16+19) /4
=44/4
=11 milli seconds (ms)

For each scheduling Gantt chart 1M,

Each average waiting time calculation 1M

6. **Attempt any TWO of the following:** **12**

a) Solve given problem by using **6M**

(i) Pre-emptive SJF

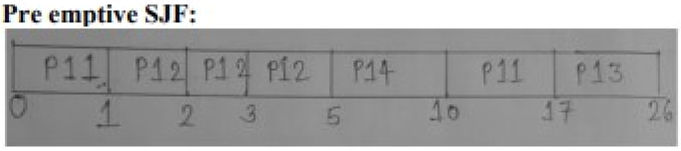
(ii) Round Robin (Time Slice = 3 ms)

Calculate average waiting time using Gantt Chart

Process	A.T.	B.T. (in ms)
P_{11}	0	8
P_{12}	1	4
P_{13}	2	9
P_{14}	3	5

Ans.

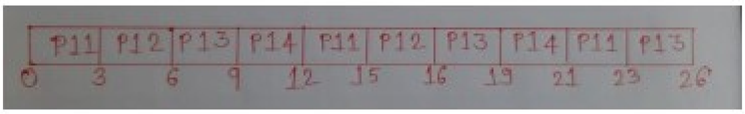
(i) Pre emptive SJF:



Waiting Time= (Total completion time –Burst time) – Arrival time

P11 –(17-8)-0 = 9ms,
P12- (5 – 4) -1 =0ms,
P13- (26-9)-2 =15ms,
P14- (10-5)-3 =2ms
Average waiting time :- (9+0+15+2)/4= 26/4=6.5 ms

(ii) Round Robin (Time Slice = 3 ms)



Waiting time: - P11 = (23-8)-0 =15ms,
P12-(16 – 4)- 1 =11ms,
P13-(26-9)-2 =15ms,
P14-(21-5)-3 =13ms
Average waiting time:- (15+11+15+13)/4=54/4= 13.5ms

Each method 3M - 1M for Gantt chart, 1M for Waiting time calculation, 1M for Average waiting time

specific user Asha.

c) **Given a page reference string with three (03) page frames. Calculate the page faults with 'Optimal' and 'LRU' page replacement algorithm respectively.**
'7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

6M

Ans. **(Representation of frame can be in any order)**

Calculate page fault with relevant diagram- 3M each

i) Optimal

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
F1	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7
F2		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1
Fault	F	F	F	F		F		F			F			F				F		

Total page faults- 9

ii) LRU

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
F1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	1	1	1	1
F2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0	0	0
F3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	7	7	7
Fault	F	F	F	F		F		F	F	F	F			F		F		F		

Total page faults-12

Fig: Context Switching

d) Ans.	**Compare Short Job First (SJF) and Shortest Remaining Time (SRTN) scheduling algorithm (any four points)**		**4M** *Any 4 correct points 1M each*

Short Job First (SJF)	**Shortest Remaining Time (SRTN)**
It is a non-preemptive algorithm.	It is a preemptive algorithm.
It involves less overhead than SRJF.	It involves more overheads than SJF.

It leads to comparatively lower throughput.	It leads to increased throughput as execution time is less.
It minimizes the average waiting time for each process.	It may or may not minimize the average waiting time for each process.
It involves lesser number of context switching.	It involves higher number of context switching.
Short processes are executed first and then followed by longer processes.	Shorter processes run fast and longer processes show poor response time.

c)	**Consider the string:** **0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7 with frame size 3 and 4, calculate page fault in both the cases using FIFO algorithm**	**6M**

Ans.

Note: Representation of frame can be in any order

FIFO :

i)Frame Size=3

FIFO implementation with frame size 3 3M

Ref.	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F3			2	2	2	1	1	1	0	0	0	3	3	3	6	6
F2		1	1	1	0	0	0	3	3	3	2	2	2	5	5	5
F1	0	0	0	3	3	3	2	2	2	1	1	1	4	4	4	7
Fault	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

OR

Ref.	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F1	0	0	0	3	3	3	2	2	2	1	1	1	4	4	4	7
F2		1	1	1	0	0	0	3	3	3	2	2	2	5	5	5
F3			2	2	2	1	1	1	0	0	0	3	3	3	6	6
Fault	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Total Page Faults :16

ii)Frame Size=4

FIFO implementation with frame size 4 3M

Ref.	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F4				3	3	3	3	3	3	3	3	3	3	3	3	7
F3			2	2	2	2	2	2	2	2	2	2	2	2	6	6
F2		1	1	1	1	1	1	1	1	1	1	1	1	5	5	5
F1	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4	4
Fault	F	F	F	F									F	F	F	F

OR

Ref.	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F1	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4	4
F2		1	1	1	1	1	1	1	1	1	1	1	1	5	5	5
F3			2	2	2	2	2	2	2	2	2	2	2	2	6	6
F4				3	3	3	3	3	3	3	3	3	3	3	3	7
Fault	F	F	F	F									F	F	F	F

Total Page Faults :08

		Total Page Faults :08	
6.		**Attempt any TWO of the following:**	**12**
	a)	**What is the average turnaround time for the following process using :** i) FCFS scheduling algorithm ii) SJF non preemptive scheduling algorithm iii) Round Robin Scheduling algorithm	**6M**

Process	Arrival time	Burst time
P_1	0	8
P_2	1	4
P_3	2	1

Ans.

Note : Any time slice can be considered by student for Round Robin e.g. 2, 3, 4 etc

For each algorithm: Gantt Chart 1M

Correct turnaround time calculation 1M each

i) **Gantt Chart FCFS**

P_1	P_2	P3
0	8	12 13

Turnaround Time =Completion Time - Arrival Time
P_1=8-0=8
P_2=12-1=11
P_3=13-2=11
Average turnaround time=Turnaround time of all processes / Number of processes
= (8+11+11) /3
=10 ms / Units

ii) **Gantt Chart SJF non-preemptive**

P_1	P_3	P_2
0	8	9 13

Turnaround Time =Completion Time - Arrival Time
P_1=8-0=8
P_2=13-1=12
P_3=9-2=7
Average turnaround time=Turnaround time of all processes / Number of processes
= (8+12+7) /3
= 9 ms/ Units

iii) <u>Gantt Chart Round Robin</u>

<u>Consider Time Quantum= 2 Units</u>

P_1	P_2	P_3	P_1	P_2	P_1	P_1

0 2 4 5 7 9 11 13

Turnaround Time =Completion Time - Arrival Time

P_1=13-0=13

P_2=9-1=8

P_3=5-2=3

Average turnaround time=Turnaround time of all processes / Number of processes

= (13+8+3) /3

= 8 ms / Units

(OR)

<u>Consider Time Quantum= 3 Units</u>

P_1	P_2	P_3	P_1	P_2	P_1

0 3 6 7 10 11 13

Turnaround Time =Completion Time - Arrival Time

P_1=13-0=13

P_2=11-1=10

P_3=7-2=5

Average turnaround time=Turnaround time of all processes / Number of processes

= (13+10+5) /3

= 9.33 ms/ Units

	b)	Explain bit map vector and linked list free space management	6M

		• Multiple threads of kernel is an overhead for operating system	
	c	**Explain LRU page replacement algorithm for following reference string. 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 Calculate the page fault.**	**6M**
	Ans	**LRU:** • The Least Recently Used (LRU) page replacement policy **replaces the page that has not been used for the longest period of time**. • LRU replacement associates with each page the time of that page's last use. • When a page must be replaced, LRU chooses the page that has not been used for the longest period of time. • The LRU policy is often used as a page-replacement algorithm and is considered to be good. • An LRU page-replacement algorithm may require substantial hardware assistance.	LRU explanation =2M Calculation =4 M

		Counters: • In the simplest case, we associate with each page-table entry a time-of-use field and add to the CPU a logical clock or counter. • The clock is incremented for every memory reference. • Whenever a reference to a page is made, the contents of the clock register are copied to the time-of-use field in the page-table entry for that page. • In this way, we always have the "time" of the last reference to each page. We replace the page with the smallest time value. **Stack:** • Another approach to implementing LRU replacement is to keep a stack of page numbers. • Whenever a page is referenced, it is removed from the stack and put on the top.	

Counters:

- In the simplest case, we associate with each page-table entry a time-of-use field and add to the CPU a logical clock or counter.
- The clock is incremented for every memory reference.
- Whenever a reference to a page is made, the contents of the clock register are copied to the time-of-use field in the page-table entry for that page.
- In this way, we always have the "time" of the last reference to each page. We replace the page with the smallest time value.

Stack:

- Another approach to implementing LRU replacement is to keep a stack of page numbers.
- Whenever a page is referenced, it is removed from the stack and put on the top.
- In this way, the most recently used page is always at the top of the stack and the least recently used page is always at the bottom.

Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1
(Frame size have not mentioned in question so assume frame size as 3 or 4)

LRU: Assume frame size=3

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
7	7	7	2		2		4	4	4	0			1		1	*	1		*
	0	0	0	*	0	*	0	0	3	3	*		3		0		0	*	
		1	1		3		3	2	2	2		*	2	*	2		7		

Page Fault=12

Assume frame size=4

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
7	7	7	7		3		3		*		*		3				7		
	0	0	0	*	0	*	0			*			0		*		0	*	
		1	1		1		4						1			*	1		*
			2		2		2	*				*	2	*			2		

Page fault=08

6		**Attempt any Two of the following:**	**12M**
	a	**The jobs are scheduled for execution as follows:**	**6M** **SJF=3 m** **FCFS=3 m** **(1m-gantt chart, 2m calculation**

Process	Arrival Time	Burst Time
P1	0	7
P2	1	4
P3	2	10
P4	3	6
P5	4	8

i) SJF
ii) FCFS
Also find average waiting time using Gantt chart.

of AWT)

Ans

Note:

SJF:

Non-Preemptive SJF

Gantt Chart:

P1	P2	P4	P5	P3	
0	7	11	17	25	35

Process	Arrival Time	Burst Time	Waiting Time
P1	0	7	0
P2	1	4	7-1=6
P3	2	10	25-2=23
P4	3	6	11-3=8
P5	4	8	17-4=13

Average waiting Time=(0+6+23+08+13)/5 = 50/5=10

OR

Preemptive SJF

Gantt Chart:

P1	P2	P1	P4	P5	P3	
0	1	5	11	17	25	35

Process	Arrival Time	Burst Time	Waiting Time
P1	0	7	0+(5-1)=4
P2	1	4	1-1=0
P3	2	10	25-2=23
P4	3	6	11-3=8
P5	4	8	17-4=13

Average Waiting Time=4+0+23+8+13/5=9.6

ii) **FCFS**

Gantt chart:

P1	P2	P3	P4	P5

0 7 11 21 27 35

Process	Arrival Time	Burst Time	Waiting Time
P1	0	7	0-0=0
P2	1	4	7-1=6
P3	2	10	11-2=9
P4	3	6	21-3=18
P5	4	8	27-4=23

Average waiting Time=0+6+9+18+23/5 =56/5=11.2

List free space management techniques? Describe any one in **6M**